Honest evaluation, and classification metrics

California housing, finished · MNIST, detecting one digit

LECTURE 3 | GÉRON CH. 2–3

Applications of Machine Learning

BSc Mathematics of Artificial Intelligence · Third year

Where we are

Last lecture, on the California housing data:

What we did	What it leaves for today
Sealed a stratified test set of 4,128 districts before looking at anything	a test set that must stay unused until the very end
Explored the 16,512 training districts, and warned against new features that are weighted sums of old ones	today shows why that warning is about linear algebra
Wrote least squares as minimising $\|\|\mathbf{X}\boldsymbol{\theta} - y\|\|_2^2$, and set its gradient to zero	the stationarity condition $\mathbf{X}^\intercal\mathbf{X}\hat{\boldsymbol{\theta}} = \mathbf{X}^\intercal y$, not yet solved
Proved the cost convex, so that stationary point is a global minimum	a minimum we now have to compute, and a model whose error we have to measure

Nothing has been fitted and nothing has been scored yet. Today does both — and then asks what a score is worth on a classification problem.

Today

1. **Finishing the housing model** — the normal equation and when it breaks; why a training score means nothing; leakage; holdout, k-fold cross-validation and pipelines; the test set, used once
2. **A classification task** — detecting one digit in MNIST, and a first detector with a headline accuracy
3. **The mathematics** — what a classifier that never fires scores, and why precision is not monotone in the threshold
4. **The method** — the confusion matrix, precision, recall and F_1, the threshold, PR against ROC, and a better model

The normal equation and its invertibility condition; leakage and why cross-validation needs a pipeline; the accuracy of the never-fires classifier; precision, recall, F_1, PR and ROC. EXAMINABLE

Open Lecture 3 in Colab

The normal equation

Last lecture's stationarity condition, $\mathbf{X}^\intercal \mathbf{X} \hat{\boldsymbol{\theta}} = \mathbf{X}^\intercal y$, rearranged:

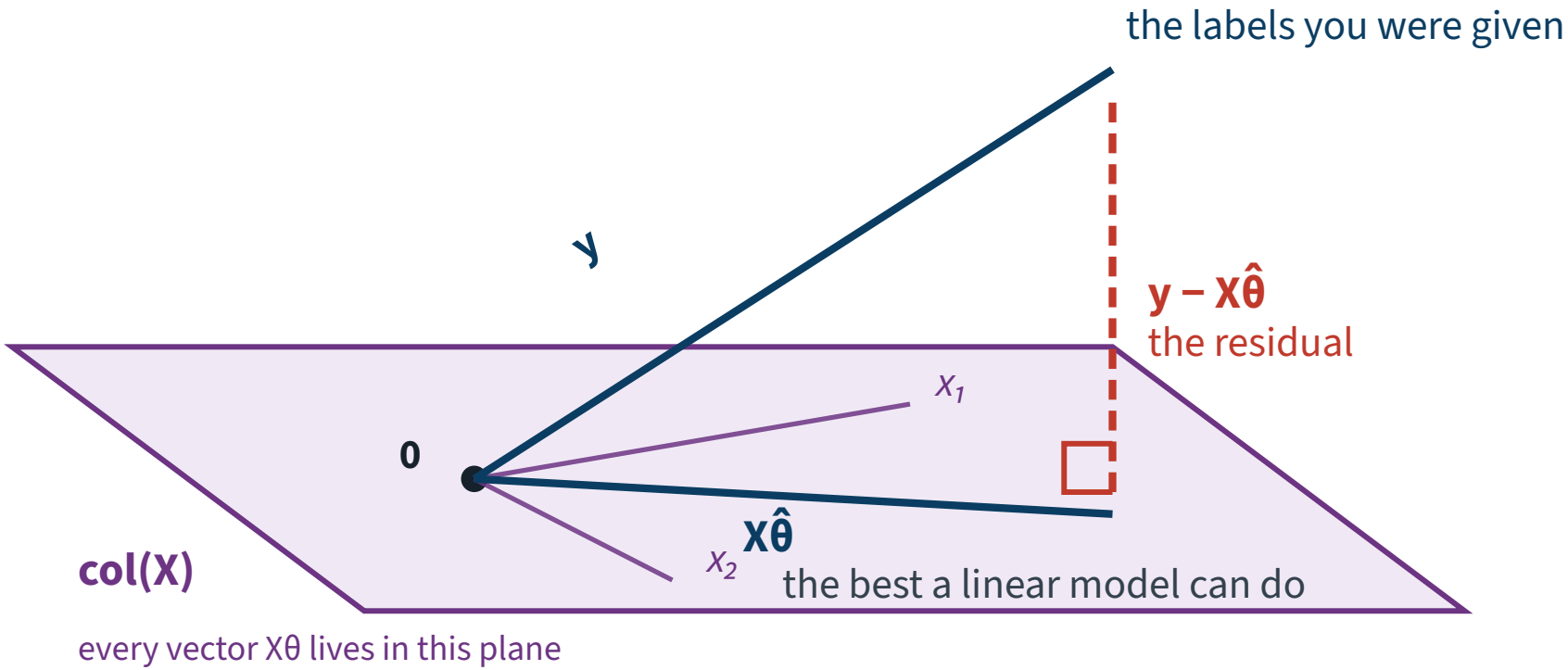
$$\hat{\boldsymbol{\theta}} = (\mathbf{X}^\intercal \mathbf{X})^{-1} \mathbf{X}^\intercal y$$

A CLOSED-FORM SOLUTION — NO ITERATION REQUIRED

A closed-form equation is composed only of a finite number of constants, variables and standard operations. No infinite sums, no limits, no integrals.

The whole thread, in one picture

$$X^\top \left(X\hat{\boldsymbol{\theta}} - \mathbf{y} \right) = \mathbf{0}$$



When $\mathbf{X}^\intercal\mathbf{X}$ is singular

The normal equation needs $(\mathbf{X}^\intercal\mathbf{X})^{-1}$, and it exists if and only if $\mathbf{X}$ has linearly independent columns. Two ways to lose that:

- **More features than instances**, $m < n + 1$: then $\operatorname{rank}(\mathbf{X}) \leq m$, and infinitely many $\boldsymbol{\theta}$ fit equally well
- **Collinear features**: one column a linear combination of others — exactly what a new feature built as a weighted sum of existing ones is. Last lecture's warning against such features was this singularity
- **Near**-collinear features leave the matrix invertible but ill-conditioned: small changes in the data move $\hat{\boldsymbol{\theta}}$ a lot

> **WHAT SCIKIT-LEARN DOES INSTEAD**
>
> It computes $\hat{\boldsymbol{\theta}} = \mathbf{X}^{+}y$ with the pseudoinverse $\mathbf{X}^{+} = \mathbf{V}\boldsymbol{\Sigma}^{+}\mathbf{U}^\intercal$ from the SVD, which is always defined: when the inverse does not exist it returns the minimum-norm solution. The fitted values are still unique; the coefficients are not.

Derive the normal equation; state and justify the invertibility condition.

EXAMINABLE

Start with a number that means nothing

Fit an unconstrained decision tree on the 16,512 training districts, then score it on those same 16,512 districts.

0.0

No error at all.

Two explanations. Only one of them is flattering.

“Then use the test set”

✗ No — not yet.

We are still choosing between models and tuning them. Every time a choice is made by looking at the test score, that score stops being an estimate of performance on unseen data and becomes a target we are fitting to.

> The test set is used **once**, at the very end, to estimate the generalisation error of a system we have already committed to.

The next slide puts this argument as an exam question. Answer 1 and 2 now; 3 and 4 need cross-validation, so they come once it exists, at the end of this half.

The same error, one step earlier

A score can also be corrupted before any model is fitted:

```
scaler = StandardScaler()
housing_scaled = scaler.fit_transform(housing_full)      # all 20,640 rows
train_set, test_set = train_test_split(housing_scaled)   # then split
```

DATA LEAKAGE — THE CONDITION

A quantity computed from rows outside the training set enters the training path. Here the mean and standard deviation were computed from a set that includes the test rows, so the model is evaluated on rows whose own values helped define the transformation applied to them.

It raises no exception and the reported score improves. How *much* it improves is not knowable from the code — sometimes nothing, sometimes everything — which is why the rule is procedural rather than a matter of judging each case.

The rule, stated once

> **NEVER VIOLATE THIS**
>
> Fit every transformer on the **training set only**. Never call `fit()` or `fit_transform()` on the validation set, the test set, or new data. Once fitted, you may `transform()` anything.

This applies to imputers, scalers, encoders, dimensionality reduction — everything that learns anything from data.

✗ We will not rely on remembering it. The next few slides build an object that makes violating it structurally impossible.

Hold out a validation set

Split the *training* set again: a smaller training set, and a **validation set** (also called the development set). Train the candidates on the reduced set, select on the validation set, retrain the winner on the full training set, and evaluate *once* on the test set.

Validation set too small

Evaluations are imprecise; you may select a suboptimal model by chance.

Validation set too large

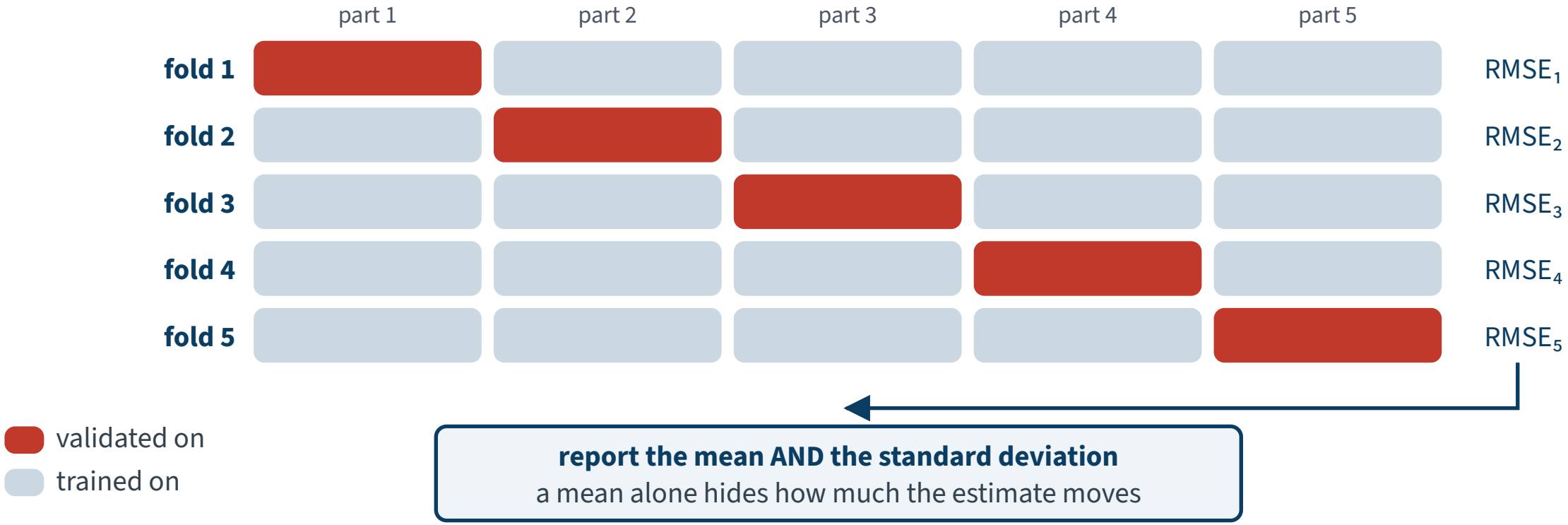
The remaining training set is much smaller than the full one — like selecting a sprinter to run a marathon.

✓ Solution: many small validation sets, used in rotation.

Every row is a complete, independent fit

The training set, partitioned five ways

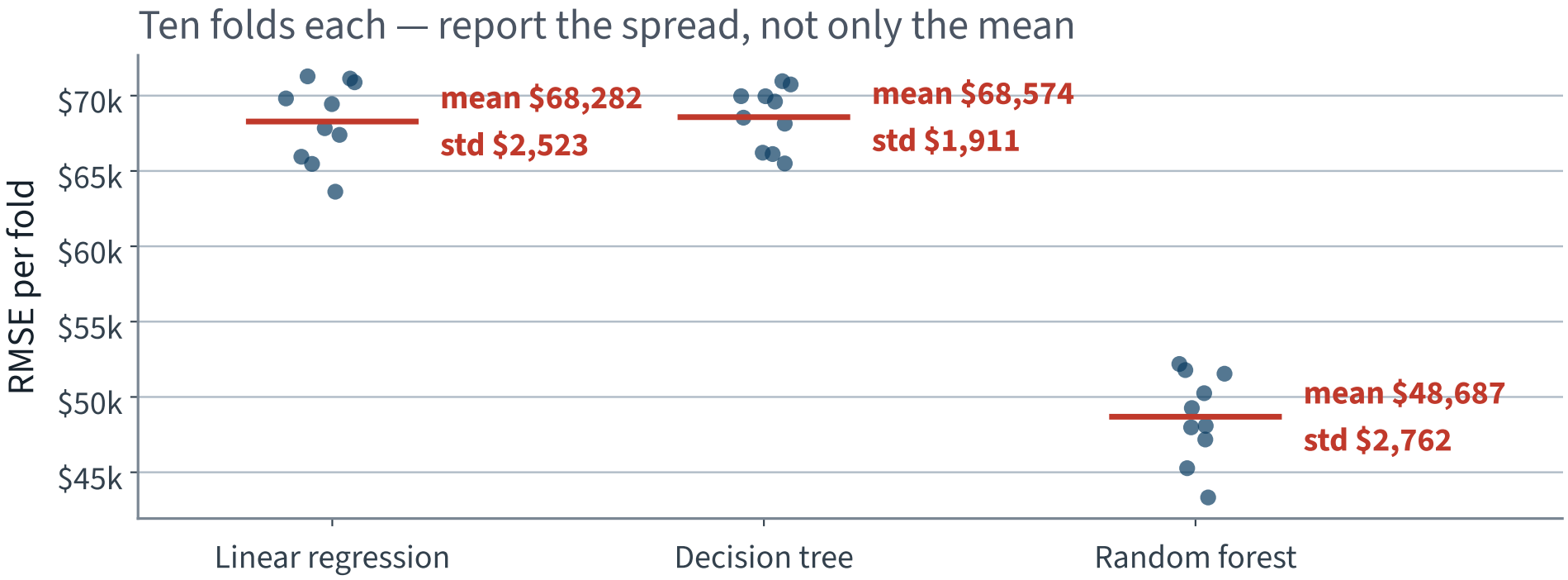
Every row is one complete fit. Every row scores on data that fit has never seen.



k = 5 drawn; the lecture uses k = 10

Five fits to get one number — and a second number, the spread, that a single holdout could never give you.

Ten folds, not one number



Linear regression, a decision tree and a random forest, each scored by ten-fold cross-validation. All three spreads are around ±$2,000–$3,000 — and that is mostly *which districts landed in the fold*, not the model. Any difference smaller than this is not a difference.

Pipelines make leakage structurally impossible

```
full_pipeline = Pipeline([
    ("prep",  preprocessing),   # imputer + scaler on numbers, one-hot on ocean_proximity
    ("model", RandomForestRegressor(random_state=42)),
])
```

When `cross_val_score` holds out fold j, the entire pipeline — imputer, encoder, scaler, model — is fitted on the other folds only.

✓ The transformers never see the held-out fold. Leakage cannot occur by accident.

Finally: the test set

```
final_model = Pipeline([
    ("prep",  preprocessing),
    ("model", RandomForestRegressor(n_estimators=200, max_features=8,
                                    random_state=42)),
]).fit(X_train, y_train)                    # the training set, and only that

final_predictions = final_model.predict(X_test)
final_rmse = root_mean_squared_error(y_test, final_predictions)
```

$49,037

The forest's two settings were chosen on validation folds of the training set before this cell ran. We `predict` on the test set; we never `fit` anything on it, and we look at it once.

You have been hired again

A national postal operator sorts handwritten postcodes automatically. A camera photographs each digit; a classifier reads it; the envelope goes down a chute.

> **THE TASK**
>
> Their audit team has found that one glyph is misread far more often than the others. They want every scanned digit that *is* a **5** pulled off the line and sent to a human verification desk.

Not “read the digit”. **Detect one digit.** Yes or no, one class against the other nine.

The data: MNIST

- 70,000 small images of handwritten digits, written by US high-school students and Census Bureau employees
- Each image is 28 × 28 pixels, flattened to **784** features
- Each feature is one pixel's intensity, an integer from 0 (white) to 255 (black)
- Every image is labelled with the digit it shows

> *The textbook calls MNIST "the hello world of machine learning". It is studied so often that it is effectively the field's control experiment: if a method cannot do something here, the something is probably not the method's fault.*

From ten classes to two

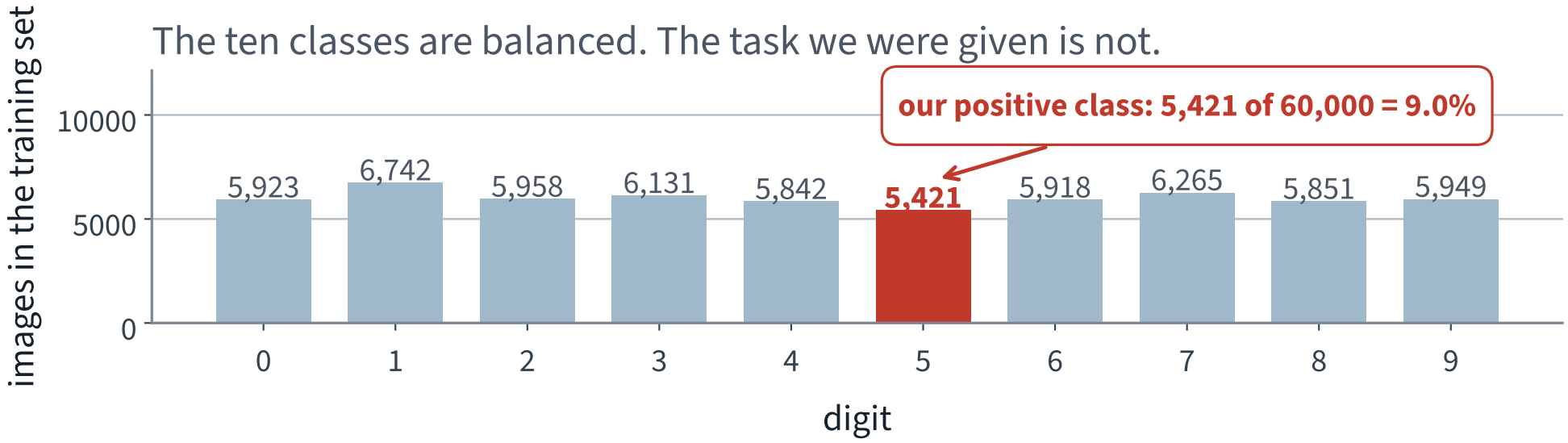
```
y_train_5 = (y_train == 5)          # True for every 5, False otherwise
y_test_5  = (y_test  == 5)

assert y_train_5.dtype == bool
assert y_train_5.sum() == 5421
print(f"{y_train_5.sum():,} positives out of {len(y_train_5):,}")
```

Two lines, and the imbalance is created. Nothing about MNIST is imbalanced; **our question is**.

✗ Assert the count. OpenML delivers the labels as strings, and a string never equals the integer 5: if they were not cast to integers first, this assertion is what tells you — it fails at `0 == 5421` instead of silently training on nothing.

Balanced data, imbalanced question



Every bar is roughly the same height. The moment we ask “is it a 5?”, one bar becomes the positive class and the other nine become the negative class.

The obvious metric for a classifier

Accuracy: the fraction of instances the classifier gets right.

$$\text{accuracy} = \frac{\#\{\, i : \hat{y}^{(i)} = y^{(i)} \,\}}{m}$$

THE PROPORTION OF CORRECT PREDICTIONS

- It is between 0 and 1, and bigger is better
- It needs no threshold, no scaling and no interpretation
- Every stakeholder in the world understands it without being taught

The headline number

```
clf = make_pipeline(StandardScaler(), SGDClassifier(random_state=42))
cv  = StratifiedKFold(n_splits=3, shuffle=True, random_state=42)
cross_val_score(clf, X_train, y_train_5, cv=cv, scoring="accuracy")
```

96.93%

Three-fold cross-validated accuracy of a linear classifier, on data the model never saw during its own fitting.

Folds: 96.85%, 96.86%, 97.09%. A spread of about two tenths of a point — the three folds agree. The scaler sits inside the pipeline, so each fold refits it on its own training part.

Before you commit: what does *nothing* score?

The cheapest possible detector. It looks at nothing and answers “not a 5” every time.

```
from sklearn.base import BaseEstimator

class NeverFires(BaseEstimator):
    def fit(self, X, y=None):
        return self
    def predict(self, X):
        return np.zeros(len(X), dtype=bool)

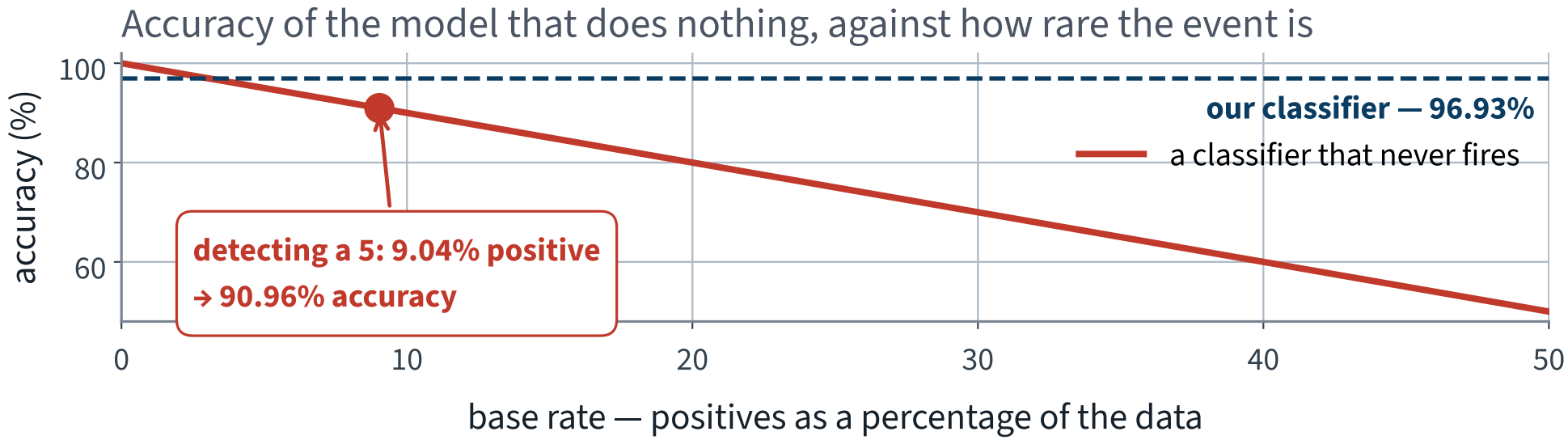
cross_val_score(NeverFires(), X_train, y_train_5, cv=cv,
                scoring="accuracy")     # array([0.90965, 0.90965, 0.90965])
```

✓ It is right 54,579 times out of 60,000 — 90.96% accuracy, with no model, no fit and no features.

Claim · the classifier that never fires

P positives, N negatives, $m = P + N$, base rate $p = P/m$. If $h_0(x) = 0$ for every x, then $\mathrm{TP} = \mathrm{FP} = 0$. Ours, $p = 0.09035$:

$$\mathrm{accuracy}(h_0) = \frac{0 + N}{m} = 1 - p = 0.90965$$



✗ An identity, not an estimate — and at a base rate of 1%, doing nothing scores 99%.

What the detector computes

Our classifier is linear. It scores a digit with $s(x) = \boldsymbol{\theta}^{\mathsf{T}}\mathbf{x}$ on the scaled pixels — regression's function — and predicts "5" when $s(x) \geq 0$.

```
clf.fit(X_train, y_train_5)
some_digit = X_train[0]                  # a training image that is a 5
score = clf.decision_function([some_digit])   # e.g. array([1231.4])
score >= 0       # array([ True]) — what predict() does
score >= 3000    # array([False]) — the same 5, missed

# one out-of-fold score per training digit, used from here on
y_scores = cross_val_predict(clf, X_train, y_train_5, cv=cv,
                             method="decision_function")
```

Regression reports the number; classification cuts it at a threshold. `predict()` hides the number and hard-codes the threshold at zero.

✓ There is no `set_threshold()` method, and there should not be. You compute the scores and you compare them yourself.

Recall is monotone; precision is not

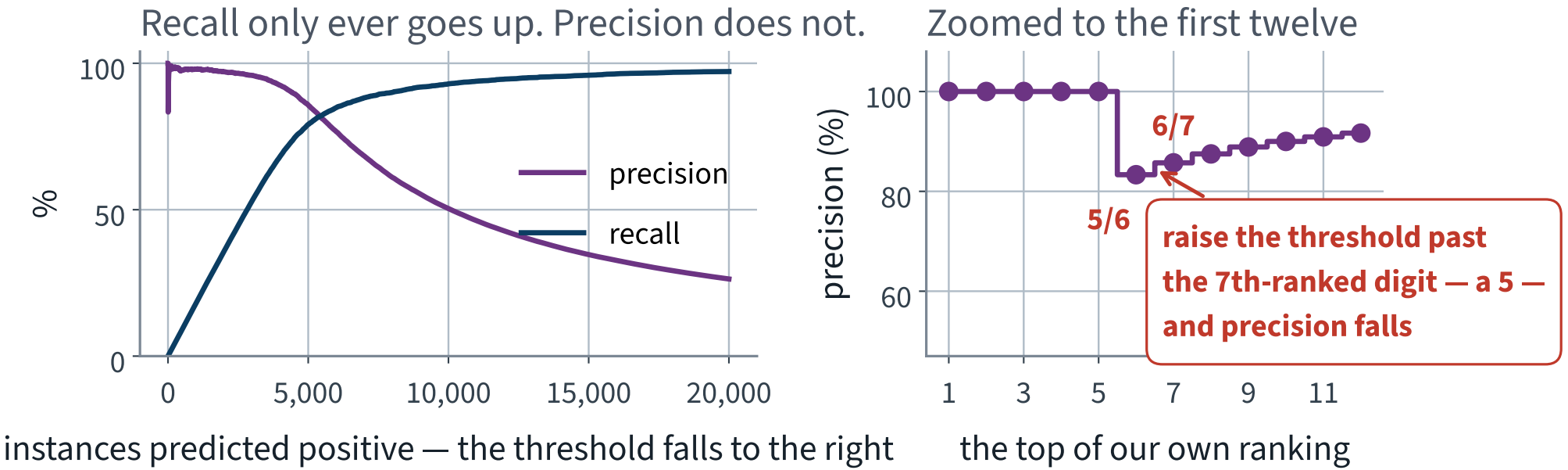
Now let the threshold move: predict positive when $s(x) \geq t$. Raising t only ever *removes* instances from the flagged set, so $\mathrm{TP}(t)$ and $\mathrm{FP}(t)$ can only fall. Recall divides $\mathrm{TP}(t)$ by P, a constant of the data, so it can only fall too.

$$\mathrm{precision}(t) \;=\; \frac{\mathrm{TP}(t)}{\mathrm{TP}(t) + \mathrm{FP}(t)}$$

Here **both** the numerator and the denominator move, in the same direction. A ratio of two decreasing quantities can do anything at all.

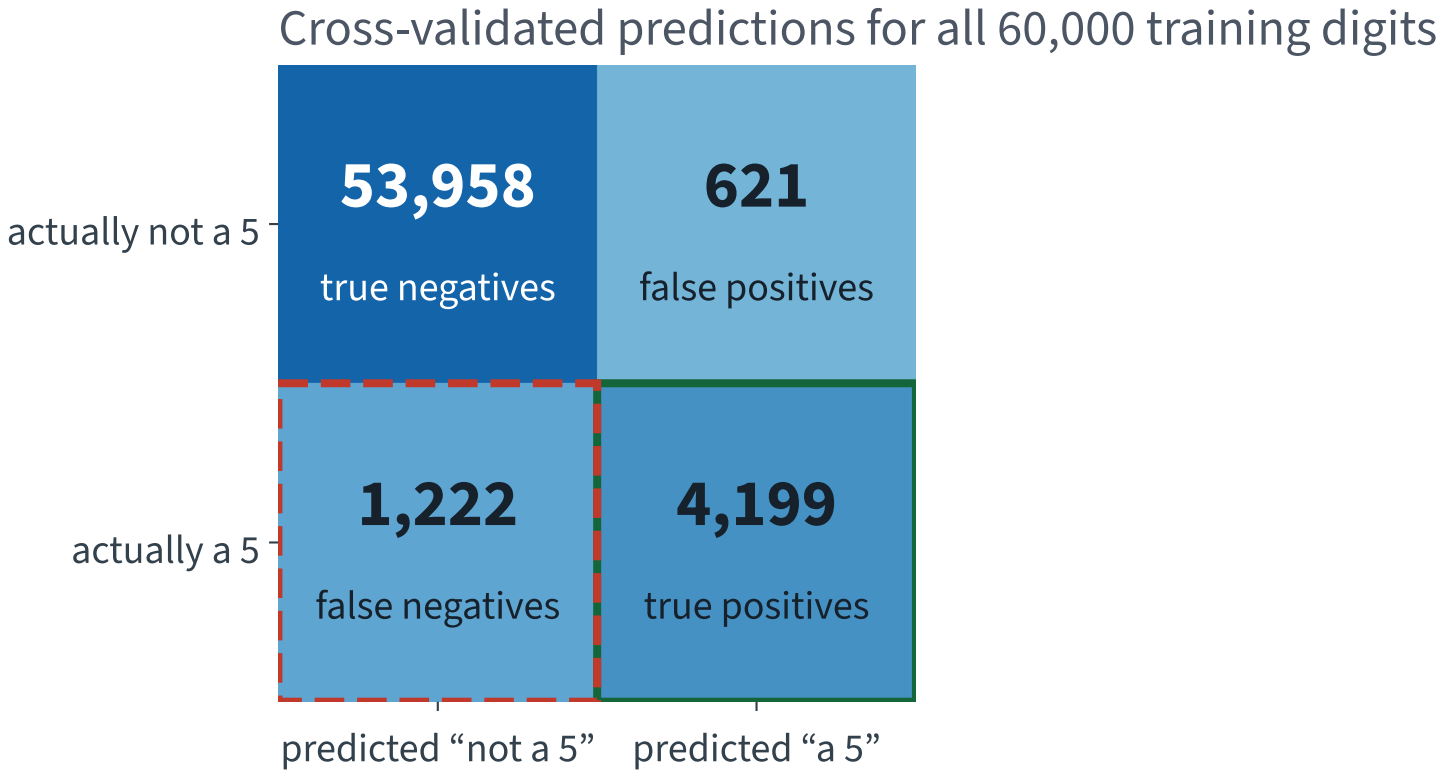
> The denominator is the size of the set *we chose to flag*. It is not a property of the data; it is a property of our decision.

Both metrics, along our own ranking



The right-hand panel zooms in on the top of our own ranking: the seventh-ranked digit is a 5, and raising the threshold past it takes precision from **6/7** down to **5/6**.

The same four numbers, laid out



✗ 1,222 fives went past unflagged — 22.5% of every 5 in the training set.

Reading a confusion matrix

- Rows are the **actual** class; columns are the **predicted** class
- The diagonal is what went right; everything off it went wrong
- A perfect classifier has zeros off the diagonal, and nothing else
- Row 1 is the negatives: 53,958 correct, 621 false alarms
- Row 2 is the positives: 1,222 missed, 4,199 caught

Both mistakes are visible, separately, and they are not the same size or the same kind. Accuracy added them together and divided by 60,000.

Ask for both numbers

```
from sklearn.metrics import precision_score, recall_score, f1_score

y_pred = (y_scores >= 0)                # what predict() would return
precision_score(y_train_5, y_pred)   # 0.8712  — 4199 / (4199 + 621)
recall_score(y_train_5, y_pred)      # 0.7746  — 4199 / (4199 + 1222)
f1_score(y_train_5, y_pred)          # 0.8200
```

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \qquad \text{recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

Recall is also called *sensitivity* or the *true positive rate*. Three names, one quantity; the third one comes back on the ROC curve later today.

One number, when you must have one

F_1 is the **harmonic** mean of precision and recall:

$$F_1 = \frac{2}{\frac{1}{\text{precision}} + \frac{1}{\text{recall}}} = 2 \cdot \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}$$

Ours is **82.00%**. The harmonic mean is pulled towards the smaller of the two numbers it combines, so a high F_1 needs both precision and recall to be high — one good number cannot carry a bad one.

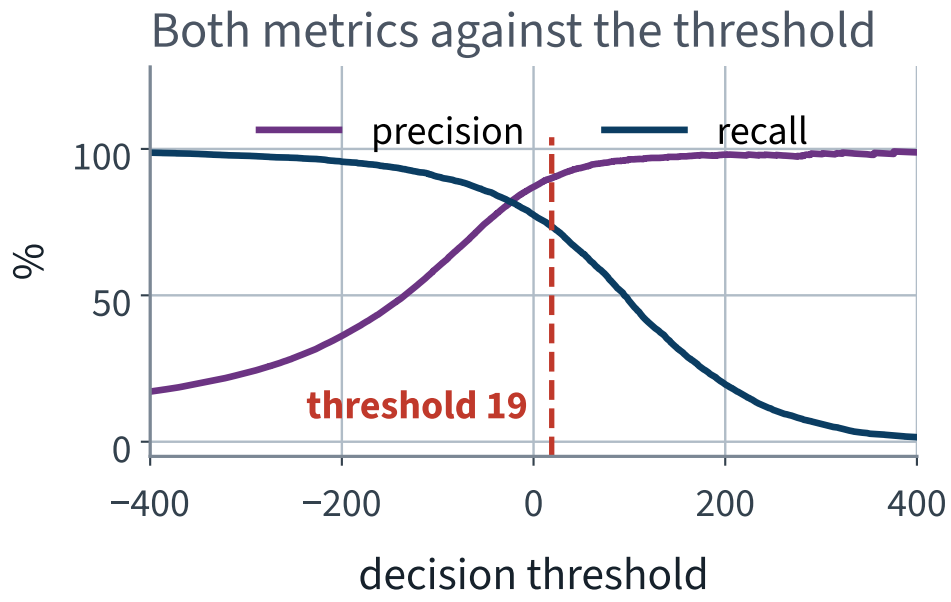
You cannot have both

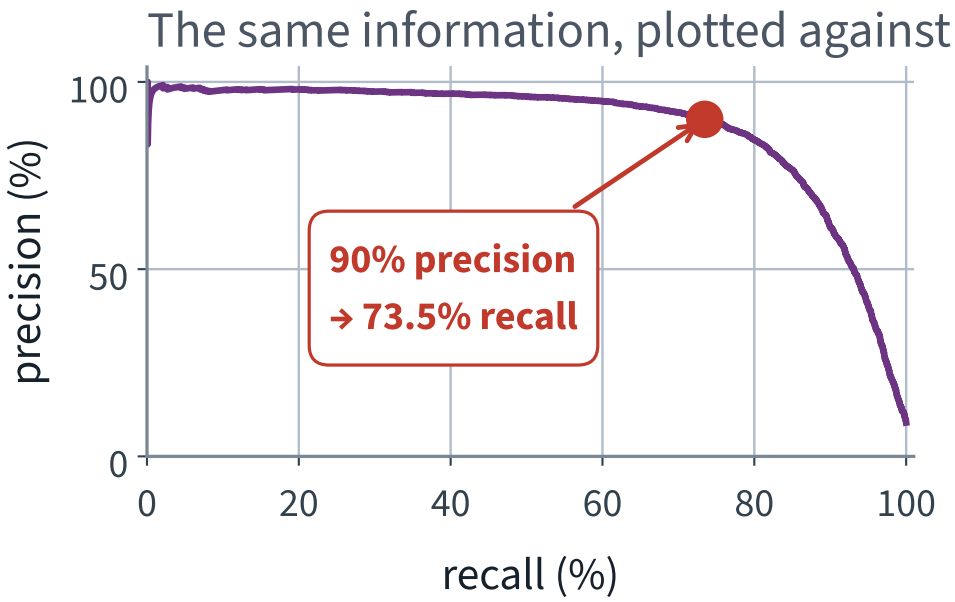
Raising recall lowers precision, and lowering recall raises it — usually.

“Usually” is doing real work in that sentence, and counting every step down our own ranking says exactly how much: precision rises at 54,579 of the steps and falls at 5,416 of them. The *trend* is reliable; the individual step is not.

> **THE PRECISION/RECALL TRADE-OFF**
>
> It is not a property of our model, or of SGD, or of MNIST. It is a property of ranking instances and cutting the ranking somewhere.

The dial, and the curve it traces





Precision starts to fall off a cliff at about 80% recall. Choose the operating point just before the cliff, not on it.

Choose the threshold deliberately

```
import numpy as np

flagged = recalls[:-1] * n_pos / precisions[:-1]   # tp / precision = tp + fp
ok  = (precisions[:-1] >= 0.90) & (flagged >= 500)  # and enough flagged
idx = int(np.argmax(ok))                           # first crossing WITH support
threshold_90 = thresholds[idx]                     # 18.95

y_pred_90 = (y_scores >= threshold_90)
print(precision_score(y_train_5, y_pred_90))   # 0.9001
print(recall_score(y_train_5, y_pred_90))      # 0.7347
```

`argmax` on a boolean array returns the first `True`. The idiom is compact and it is not obvious; read it once and remember it.

✗ Ninety per cent precision costs us 3.98 points of recall, from 77.46% down to 73.47%.

The ROC curve

Same sweep, different pair of axes: the true positive rate against the false positive rate.

$$\mathrm{TPR} = \frac{\mathrm{TP}}{P} = \text{recall}, \qquad \mathrm{FPR} = \frac{\mathrm{FP}}{N} = 1 - \text{specificity}$$

```
from sklearn.metrics import roc_curve, roc_auc_score
fpr, tpr, thresholds = roc_curve(y_train_5, y_scores)
roc_auc_score(y_train_5, y_scores)          # 0.9723
```

Note that **both** denominators are fixed by the data. Neither axis moves when we change the threshold's effect on the flagged count — which is precisely why the ROC curve behaves so much better than the PR curve, and why that is a warning rather than a recommendation.

Which curve to prefer

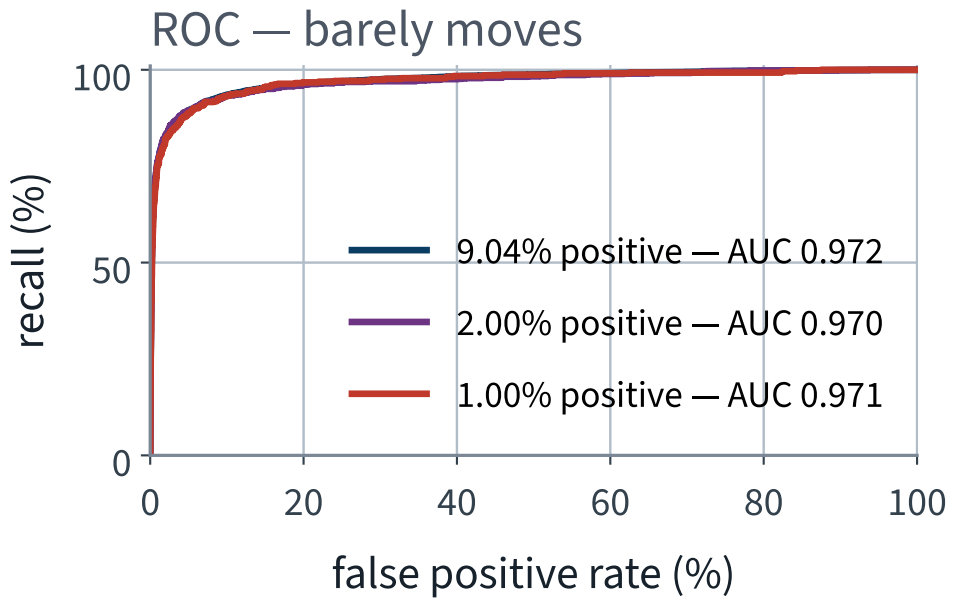
> THE RULE
>
> *Prefer the precision/recall curve whenever the **positive class is rare**, or whenever you care more about false positives than about false negatives. Otherwise use the ROC curve.*

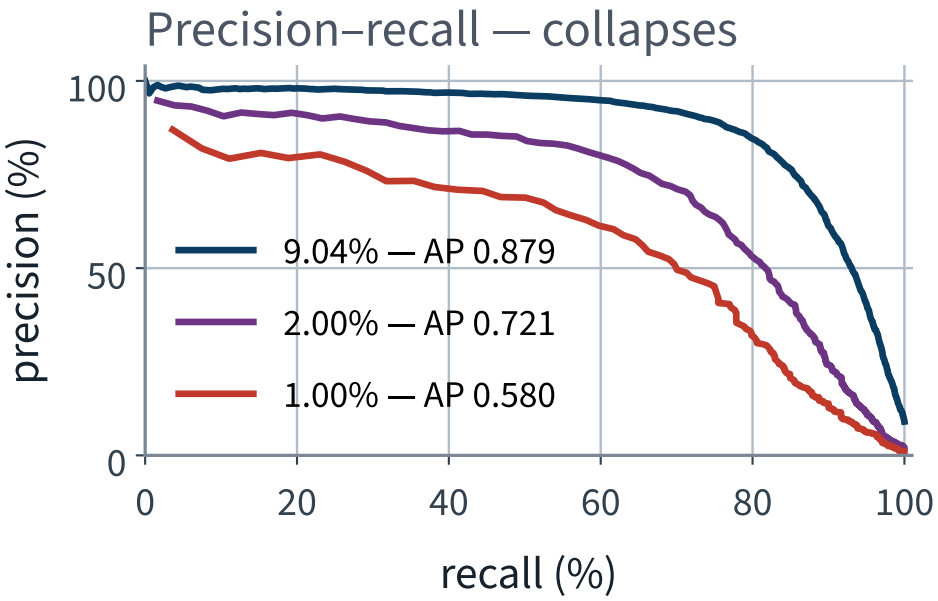
The reason is arithmetic. FPR divides false positives by N, which is huge when positives are rare, so a large number of false alarms produces a small, comfortable-looking FPR. Precision divides the same false positives by the flagged count, which is small.

✓ Do not take that on trust. Measure it.

ROC on a rare positive class

Our scores, unchanged; only the positives are thinned, so the base rate falls from 9.04% to 2% and then 1%.





FAILURE CONDITION

Three indistinguishable ROC curves, three very different PR curves, from one set of scores: the false-positive rate divides by the abundant class, so ROC cannot see what a rare positive costs.

A better classifier, on the same metrics

```
forest = RandomForestClassifier(
    n_estimators=100, random_state=42)
y_proba = cross_val_predict(
    forest, X_train, y_train_5, cv=cv,
    method="predict_proba")
y_scores_forest = y_proba[:, 1]
```

A forest has no `decision_function`. The positive-class column of `predict_proba` plays the role of the score, and every curve works unchanged.

Metric	SGD	Random forest
Accuracy	96.93%	98.77%
Precision	87.12%	98.93%
Recall	77.46%	**87.31% ✓**
F_1	82.00%	92.76%
ROC AUC	0.9723	0.9984
Average precision	0.8788	0.9883
Recall available at 90% precision	73.47%	**97.51% ✓**

Accuracy improves by 1.84 points, which reads as polish. The last row improves by **24.04 points**, and it is the row the brief is about.

The notebook for this lecture

Open Lecture 3 in Colab

- **Runs on free CPU**
- Part 1, on the housing data: the normal equation with the residual checked orthogonal to every column, the training score of zero, ten-fold cross-validation of three models inside one pipeline, and the single use of the test set
- Part 2, on MNIST: the never-fires baseline, the detector, the confusion matrix, precision, recall and F_1, the curves, the threshold choice, and the forest — with the never-fires accuracy checked against $1 - p$, and the non-monotonicity of precision checked at every one of the 59,999 thresholds

> **WHAT TO CHANGE BEFORE THE NEXT LECTURE**
>
> Detect a different digit. Change one character — the `5` in `y_train == 5` — and re-run. The baseline moves, the accuracy moves with it, and the pair tells you something neither does alone.

What we did

1. Solved least squares in closed form, $\hat{\boldsymbol{\theta}} = (\mathbf{X}^\intercal \mathbf{X})^{-1} \mathbf{X}^\intercal y$, read it as an orthogonal projection, and saw that it breaks when the columns of $\mathbf{X}$ are dependent — which the SVD pseudoinverse survives
2. Saw a training error of zero mean nothing, and a leak improve a score silently: every transformer is fitted on training data only, and a `Pipeline` inside cross-validation enforces it
3. Measured with k-fold cross-validation, reported the fold spread with the mean, and touched the test set once
4. Framed a detection task as binary classification: the positive class is rare by construction, and the classifier that never fires scores exactly $1 - p$
5. Showed that recall is monotone in the threshold and precision is not
6. Read the confusion matrix, used precision, recall and F_1, chose a threshold deliberately, and preferred the PR curve to ROC because the positives are rare

NOT PRESENTED — FOR AFTERWARDS

Exercises Lecture 3

five, in the style of the written paper

Exercises — Lecture 3

1 The normal equation is $\hat{\theta} = (X^\mathsf{T} X)^{-1} X^\mathsf{T} y$. Your training matrix X has a column equal to the sum of two others. State what happens to $X^\mathsf{T} X$, and what the pseudoinverse X^{+} returns instead. [5]

2 A pipeline scales the features and then fits a model, and is passed whole to `cross_val_score`. Explain what would go wrong if the scaling were done once, before the call. [5]

3 On the MNIST 5-detector, a classifier that never fires scores 90.96% accuracy on the training set. State what that number is a measurement of. [3]

Solutions on Lecture 4's deck.

Exercises — Lecture 3 (continued)

4 Prove or disprove: as the decision threshold is lowered, precision increases monotonically. [5]

5 Two models have the same ROC AUC. One is far better on a precision–recall curve. What does that tell you about the problem? [4]

Solutions on Lecture 4's deck.

THE FIVE SET AT THE END OF LECTURE 2

Solutions Lecture 2

not part of this lecture

Solutions — Lecture 2

1. A colleague reports their model's error on the California housing test set, and says they chose the…

✓ It estimates the best-of-several error on that particular test set; it is not an estimate of the generalisation error, the error on new districts.

- Choosing the minimum of several numbers measured on the same data makes that minimum a *selection*, not a measurement
- This is data snooping: the test set picked the model, so its error is an optimistic estimate of generalisation error, which needs data that took part in no choice

2. You are given a stratified split on income category and a naive random split of the same data. A model scored…

✓ No.

- A similar error on *one* draw says nothing about the spread over draws, and the stratification is there to reduce that spread
- The failure a naive split risks is sampling bias — measured here as up to 6.4% off in the income mix, against 0.36% stratified — and a single comparison of errors cannot show it

Solutions — Lecture 2 (2)

3. On the training set, longitude has a correlation of $r = -0.05$ with the median house value. A colleague…

✓ r measures only a linear relationship, and price depends on location non-linearly. The geographic scatter coloured by price refutes it.

- $r \approx 0$ rules out a straight-line trend in longitude alone, not dependence
- Prices are highest along the coast and around the two big metropolitan areas, a pattern no single coordinate traces as a straight line

4. The course reports the RMSE of a regression model, yet training minimises the MSE. Explain why the two give…

✓ The square root is strictly increasing on $[0, \infty)$, so $\arg\min \sqrt{f} = \arg\min f$. The RMSE is reported because it is in the units of the target, dollars.

- A strictly increasing function preserves the order of the values, so it preserves where the minimum is
- Minimising the MSE avoids differentiating a square root: same minimiser, simpler algebra

Solutions — Lecture 2 (3)

5. The Hessian of the MSE of a linear model is $\frac{2}{m}\mathbf{X}^\intercal\mathbf{X}$. Show that it is…

✓ For any $\mathbf{v}$, $\mathbf{v}^\intercal\mathbf{X}^\intercal\mathbf{X}\mathbf{v} = \|\mathbf{X}\mathbf{v}\|_2^2 \geq 0$. So the MSE is convex, and any stationary point is a global minimum.

- The quadratic form is a squared norm, so it cannot be negative, whatever $\mathbf{X}$ is
- A convex function has no local minima other than the global one, so setting the gradient to zero finds *the* minimum

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Training models

Gradient descent, derived — the update rule, the step size, why the features must be scaled, and one instance at a time

Logistic regression on the Titanic passengers: a probability, not a label, and coefficients read as log-odds

Calibration, and softmax regression for more than two classes

Run the Lecture 3 notebook first